

Machine Learning Research Proposal

Smart Grid & Energy (Green AI)

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1 Introduction

The energy sector is changing fast. Instead of relying on large, centralized power plants, it is moving toward decentralized systems known as Smart Grids. This shift is happening because of growing concerns about sustainability, the increasing use of renewable energy sources such as solar and wind, and the need for a more reliable power supply.

Unlike traditional grids, Smart Grids depend heavily on real-time information. Smart meters, sensors, and phasor measurement units (PMUs) continuously collect and share data, resulting in a huge volume of information that helps operators monitor, control, and optimize the power system more efficiently.

Our area of interest lies in applying **Green AI and Deep Learning** to solve critical challenges in this domain. Specifically, we are motivated by the difficulty of managing grid stability in the face of intermittent renewable energy and the complex, non-linear nature of modern power distribution. The following proposals explore three distinct approaches: structural analysis using Graph Neural Networks, predictive modeling for renewables using Hybrid Architectures, and control optimization using Reinforcement Learning.

2 Proposal Options

Note: Multiple proposal ideas are included below. One proposal will be finalized later.

Proposal Option 1: Graph Neural Networks (GNN) for Fault Detection and Localization in Power Grids

Problem Statement:

Power grids are complex, interconnected graph structures. Traditional fault detection

methods often rely on threshold-based rules or treat grid data as simple tabular data, ignoring the physical topology (connections between transformers and lines). This leads to slow or inaccurate fault localization, increasing the risk of cascading failures (blackouts).

Objectives:

- To model the power grid network topology as a graph structure where buses are nodes and lines are edges.
- To develop a Spatial-Temporal Graph Neural Network (ST-GNN) that learns dependencies from both grid topology and time-series sensor data.
- To achieve high accuracy in detecting and pinpointing specific nodes (buses) where faults occur in real-time.

Proposed Methodology:

We will implement a **Graph Convolutional Network (GCN)** combined with an **LSTM** layer (ST-GCN). The GCN will extract spatial features from the grid topology (adjacency matrix), while the LSTM will capture temporal dependencies in voltage and frequency fluctuations. The model will classify the state of the grid as “Normal” or “Faulted” and identify the specific node index of the fault.

Dataset:

- **Source:** IEEE 14-bus, 39-bus, or 118-bus standard test systems (simulated data).
- **Size:** $\approx 10,000+$ simulated time-step samples.
- **Type:** Structured Time-Series and Graph Adjacency Matrix.

Evaluation Metrics:

- Detection Accuracy (%)
- Precision, Recall, and F1-Score
- Localization Error (average distance from actual fault)

Expected Outcome:

A robust framework that outperforms traditional CNN/ANN baselines by effectively utilizing the graph structure of the grid.

Proposal Option 2: Short-Term Solar Power Forecasting using Hybrid CNN-Transformer Models

Problem Statement:

The integration of solar energy into the grid is hindered by its intermittency—clouds and weather changes cause rapid fluctuations in power generation. Inaccurate forecasting forces grid operators to keep fossil-fuel backup generators running. Existing LSTM models often struggle with long-term dependencies in weather patterns.

Objectives:

- To preprocess multi-variate weather data (temperature, irradiance, humidity, wind speed) for time-series modeling.

- To design a hybrid model where CNN layers extract local trend features and a Transformer model (using Self-Attention) captures long-term dependencies.
- To reduce the prediction error for 1-hour to 24-hour ahead forecasting horizons.

Proposed Methodology:

We propose a **Hybrid CNN-Transformer** architecture. 1D-CNN layers will be used first to extract short-term local features and reduce noise from raw weather sensor data. The output will be fed into a Transformer Encoder block to learn global temporal correlations via the attention mechanism.

Dataset:

- **Source:** NREL (National Renewable Energy Laboratory) Solar Power Data.
- **Size:** 1 to 3 years of hourly or 5-minute interval data.
- **Type:** Multivariate Time-Series (Numerical).

Evaluation Metrics:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- R-squared (R^2) Coefficient

Expected Outcome:

A highly accurate forecasting model that demonstrates lower RMSE compared to standard LSTM or ARIMA models.

Proposal Option 3: Deep Reinforcement Learning (DRL) for Demand Response Management

Problem Statement:

Balancing energy supply and demand in real-time is challenging. Automating “Demand Response” programs for millions of devices without compromising user comfort is a complex optimization problem that traditional rule-based controllers cannot handle efficiently.

Objectives:

- To formulate the Demand Response problem as a Markov Decision Process (MDP).
- To train a DRL agent that dynamically adjusts energy prices or controls appliance usage to flatten the peak load curve.

Proposed Methodology:

We will utilize **Proximal Policy Optimization (PPO)** or **Deep Q-Networks (DQN)**. The agent will observe the grid state and take actions (price adjustment) to maximize a reward function based on stability and user comfort.

Dataset:

- **Source:** Pecan Street Dataport or OpenAI Gym / Grid2Op environments.
- **Type:** Reinforcement Learning Environment (State-Action pairs).

Evaluation Metrics:

- Cumulative Reward
- Peak-to-Average Ratio (PAR) reduction

3 Tools and Technologies

- **Programming Language:** Python 3.9+
- **Frameworks:** PyTorch (Geometric), TensorFlow, Scikit-learn
- **Libraries:** Pandas, NumPy, NetworkX, Matplotlib

4 References

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4. Kipf, T.N. and Welling, M., “Semi-Supervised Classification with Graph Convolutional Networks,” *arXiv preprint arXiv:1609.02907*, 2016.